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usepackage[a4paper,margin=0.65in]geometry
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Collected Mathematical Papers of Arthur Cayley  
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(Papers 913, 914, 915)

913.

ON THE EPITROCHOID.

[From the *Messenger of Mathematics*, vol. xx. (1891), pp. 150–158.]

IF we have a curve  $C_1$  rolling on a fixed curve  $C$ , and consider the epitrochoid described by a point  $P_1$  attached to and carried along with the curve  $C_1$ , then there is a known construction for the radius of curvature at any point of the epitrochoid; the construction is probably much older, but I refer to Mannheim, *Géométrie Descriptive*, Paris, 1886, pp. 177 and 194. In fact, if  $Q$  be a position of the point of contact,  $P_1$  the corresponding position of the describing point,  $R_1$  and  $R$  the radii of curvature at  $Q$  of the two curves respectively,  $\rho$  the distance  $QP_1$ , and  $\phi$  the inclination of this distance to the common normal at  $Q$ , then the radius of curvature  $x$  at the point  $P_1$  of the epitrochoid is given by the formula

$$\frac{x}{x - \rho} = \frac{\rho}{\cos \phi} \left( \frac{1}{R} + \frac{1}{R_1} \right).$$

I prove this as follows: take  $Q, Q'$  consecutive positions of the point of contact,  $P_1, P'_1$  the corresponding positions of  $P_1$ ; the centre of curvature is the intersection of the lines  $P_1Q, P'_1Q'$ ; hence, if for a moment  $M$  and  $N$  are the perpendicular distances between these two lines at the points  $P_1$  and  $Q$  respectively, we have  $\frac{x}{x - \rho} = \frac{M}{N}$ . Let  $ds$  be the element  $QQ'$  considered as belonging to each of the two curves respectively,  $\theta$  and  $\theta_1$  the angles which

this element subtends at the two centres of curvature respectively; we have  $ds = R\theta = R_1\theta_1$ , whence

$$\theta + \theta_1 = ds \left( \frac{1}{R} + \frac{1}{R_1} \right).$$

The instantaneous centre is  $P_1$ , and hence

$$M = P_1P'_1 = \rho(\theta + \theta_1); \quad \text{also } N = ds \cos \phi;$$

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we thus have

$$\frac{x}{x - \rho} = \frac{\rho(\theta + \theta_1)}{ds \cos \phi};$$

or, substituting for  $\theta + \theta_1$  the foregoing value, we have the required expression

$$\frac{x}{x - \rho} = \frac{\rho}{\cos \phi} \left( \frac{1}{R} + \frac{1}{R_1} \right).$$

If the curve  $C_1$  roll on a straight line,  $R = \infty$ , and therefore

$$\frac{x}{x - \rho} = \frac{\rho}{R_1 \cos \phi}.$$

Suppose that at any instant we have on the epitrochoid an inflexion, then  $x = \infty$ , and we have  $\rho = R_1 \cos \phi$ ; this denotes that  $P_1$  is a point of the circle described on the radius of curvature at  $Q$  as its diameter, and we have thus the theorem:

*If at any instant we consider the circle, which has for its diameter the radius of curvature of the rolling curve at its point of contact with the right line, then each point on this circle is an inflexion on the epitrochoid described by the same point as the curve rolls on the straight line.*

The circle in question for any point  $Q$  of the rolling curve  $C_1$  is of course a circle, having its centre on the normal at  $Q$  and its radius equal to one-half of the radius of curvature; we may call it the *circle of double curvature*. Regarding  $Q$  as a given point on the curve  $C_1$ , and supposing that the circle of double curvature intercepts the curve  $C_1$  in a point  $X$ , then taking  $P_1$  at  $X$ , we have the theorem:

*If the curve  $C_1$  roll on a straight line, then the epitrochoid described by the point  $X$  has an inflexion corresponding to the position of the rolling curve for which the point  $Q$  comes into contact with the straight line.*

Consider the curve  $C_1$  referred to axes fixed in the plane of the curve and moveable with it; take  $x, y$  as the coordinates of the point  $Q$ ;  $\alpha, \beta$  as the coordinates of the corresponding centre of curvature, and  $X, Y$  as current coordinates; the equation of the circle of double curvature is

$$(X - x)(X - \alpha) + (Y - y)(Y - \beta) = 0,$$

viz. this is the equation of a circle having for a diameter the two points  $(x, y)$  and  $(\alpha, \beta)$ .

Suppose that  $C_1$  is a conic; the circle of double curvature at any point  $Q$  meets it in the point  $Q$  counting twice, and in two other points, say  $P_1$ ,

$P_2$ , which are determined as the intersections of the conic by the line  $P_1P_2$  which joins them. I propose to determine the equation of this line. The investigation would be more symmetric for the ellipse, but I nevertheless prefer to consider for the curve  $C_1$  the hyperbola  $b^2x^2 - a^2Y^2 - a^2b^2 = 0$ .

Representing for a moment the equation of the circle by  $U = 0$ , and that of the hyperbola by  $V = 0$ , it must be possible to determine the ratio of the coefficients  $p, q$  in such wise that  $pU + qV = 0$  shall break up into a pair of lines, one of which is the tangent at  $Q$  of the hyperbola; and then the other of them will be the required line  $P_1P_2$ .

The equation of the tangent is  $b^2xX - a^2yY - a^2b^2 = 0$ ; we then see that the equation of the line  $P_1P_2$  must be of the form  $b^2xX + a^2yY + \Omega = 0$ ; and we have to find  $p, q, \Omega$  so as to verify the identity

$$(p + qb^2)X^2 + (p - qa^2)Y^2 - p(\alpha + x)X - p(\beta + y)Y + p(\alpha x + \beta y) - qa^2b^2 \\ = (b^2xX - a^2yY - a^2b^2)(b^2xX + a^2yY + \Omega);$$

that is, we ought to have

$$\begin{aligned} p + qb^2 &= b^4x^2, \\ p - qa^2 &= a^4y^2, \\ p(x + \alpha) &= (\Omega - a^2b^2)b^2x, \\ qa^2b^2 - p(\alpha x + \beta y) &= \Omega a^2b^2. \end{aligned}$$

It will be recollected that  $\alpha, \beta$  denote the coordinates of the centre of curvature corresponding to the point  $(x, y)$  of the hyperbola; their values thus are

$$\alpha = (a^2 + b^2) \frac{x^3}{a^4}, \quad \beta = -(a^2 + b^2) \frac{y^3}{b^4}.$$

There is no difficulty in finding the values

$$p = \frac{a^4b^4}{a^2 + b^2}, \quad q = \frac{b^4x^2 + a^4y^2}{a^2 + b^2};$$

whence also

$$\begin{aligned} q - a^2b^2 &= \frac{a^4b^4}{a^2 + b^2} \left\{ \left( \frac{1}{a^2} + \frac{1}{b^2} \right) y^2 - 1 \right\}, \\ q + a^2b^2 &= \frac{a^4b^4}{a^2 + b^2} \left\{ \left( \frac{1}{a^2} + \frac{1}{b^2} \right) x^2 + 1 \right\}, \\ \Omega &= -a^2b^2 \frac{x^2 + y^2}{a^2 + b^2}; \end{aligned}$$

these satisfy the identities in question.

The equation of the line  $P_1P_2$  is thus

$$b^2xX + a^2yY - \frac{a^2b^2}{a^2 + b^2}(x^2 + y^2) = 0,$$

and combining this with the equation

$$b^2X^2 - a^2Y^2 - a^2b^2 = 0$$

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of the hyperbola, we find

$$\left\{ xX - \frac{a^2}{a^2 + b^2}(x^2 + y^2) \right\}^2 = \frac{a^4 y^2 Y^2}{b^4}, = \frac{a^2 y^2}{b^2}(X^2 - a^2),$$

that is,

$$X^2 \left( x^2 - \frac{a^2 y^2}{b^2} \right) - \frac{2a^2 x}{a^2 + b^2}(x^2 + y^2) X + \frac{a^4}{(a^2 + b^2)^2}(x^2 + y^2)^2 + \frac{a^4 y^2}{b^2} = 0,$$

viz.

$$X^2 - 2 \frac{x^2 + y^2}{a^2 + b^2} xX + a^2 \left\{ \left( \frac{x^2 + y^2}{a^2 + b^2} \right)^2 + \frac{y^2}{b^2} \right\} = 0,$$

and we thence obtain

$$X = \frac{x^2 + y^2}{a^2 + b^2} x \pm \frac{ay}{b} \sqrt{\left\{ \left( \frac{x^2 + y^2}{a^2 + b^2} \right)^2 - 1 \right\}},$$

$$Y = -\frac{x^2 + y^2}{a^2 + b^2} y \pm \frac{bx}{a} \sqrt{\left\{ \left( \frac{x^2 + y^2}{a^2 + b^2} \right)^2 - 1 \right\}},$$

values belonging to the points  $P_1, P_2$  respectively: these points are thus real or imaginary, according as  $x^2 + y^2$  is greater or less than  $a^2 + b^2$ . In the limiting case  $x^2 + y^2 = a^2 + b^2$ , we have  $X = x, Y = -y$ , viz. here the circle of double curvature has its centre on the axis of  $x$ , and has double contact with the hyperbola at the points  $(x, y)$  and  $(x, -y)$  respectively.

Suppose, to fix the ideas,  $x$  and  $y$  are both positive, i.e. that  $Q$  is on the right-hand upper half branch of the hyperbola; and that  $x^2 + y^2 =$  or  $> a^2 + b^2$ , so that  $P_1, P_2$  are both real. The two values of  $X$  are both positive, those of  $Y$  are originally both negative, and as  $Q$  moves along the half branch of the hyperbola one of these values increases negatively, the other increases positively until it becomes  $= 0$ , after which it becomes positive and continually increases: say the points  $P_1, P_2$  are originally on the right-hand lower half branch, but one of them, say  $P_1$ , moves up to the vertex, and we afterwards have  $P_1$  on the upper half branch and  $P_2$  on the lower half branch. The limiting case is when  $P_1$  is at the vertex; here

$$\frac{x^2 + y^2}{a^2 + b^2} = \frac{a}{x},$$



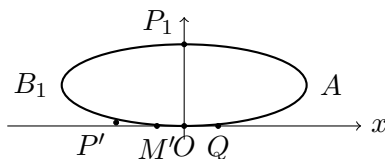
giving for  $(x, y)$  a real point, which is easily constructed on the upper half branch. There is thus on the half branch a real position of  $Q$ , such that the corresponding circle of double curvature passes through the vertex of the same half branch.

Hence in the epitrochoid described by the vertex of a hyperbola rolling on a right line there is always an inflexion.

Imagine the branch  $B'AB$  of a hyperbola rolling on the line  $y = 0$ , where  $A$  is the vertex,  $B', B$  the points at infinity, and so that as  $x$  increases positively (that is, towards the right) the point of contact  $Q$  passes from  $A$  towards  $B$ ; consider the epitrochoid described by the vertex  $A$ . Supposing that the point of contact is at first at  $A$ ; the point  $A$  is a cusp on the epitrochoid, the motion of  $A$  is at first vertically upwards, and towards the right, the curve being concave to the axis; we then come to a position of the point of contact for which there is an inflexion; the motion of  $A$  is still upwards and towards the right, but the curve has now

become convex to the axis, and the motion continues in this manner as the point of contact moves along the hyperbola towards  $B$ ; when the point of contact is at  $B$ , i.e. when the asymptote has come to coincide with the axis, then the motion of  $A$  is again vertically upwards; the other half branch of the hyperbola now comes into contact with the axis; the point  $A$  still moves upwards but towards the left, the curve being concave to the axis, and this continues until the point of contact arrives at the other vertex (i.e. the vertex on the other branch), when the motion of  $A$  is horizontal and towards the left, viz. the coordinate  $y$  has here a maximum value, equal transverse axis of the hyperbola; the motion of  $A$  is thenceforth towards the left and downwards, until the other asymptote comes to coincide with the curve, after which it is downwards and towards the right (the point of contact being now on the half  $B'A$  of the original branch  $BAB'$ ), and ultimately the point of contact is again at  $A$ , and we have  $y = 0$ , a cusp; the curve is symmetrical in regard to the before-mentioned maximum ordinate: it is to be noticed that the distance between the two cusps is equal to four times the difference between the length of the half asymptote and the half branch of the hyperbola (this difference being a finite value expressible by elliptic functions), and that, as already mentioned, the maximum ordinate is equal to the transverse axis. The above described portion of the curve is of course repeated continually right and left *ad infinitum*; two consecutive portions intersect each other, and there is thus a real node vertically above each cusp of the infinite curve. We have, in what precedes, a complete explanation of the motion of a hyperbola rolling on a straight line, and in particular our explanation of the form of the epitrochoid described by a vertex; the epitrochoid described by any other point of the curve is of a similar character but must be non-symmetrical in regard to the maximum ordinate; and it would not be difficult to explain the form of the curve for other positions of the describing point; but as regards the analytical theory, I prefer to consider the case of the ellipse.

In the case of the ellipse  $b^2X^2 + a^2Y^2 - a^2b^2 = 0$  rolling on the line  $y = 0$  (see figure), take as origin the point  $O$  which comes in contact with the extremity  $B$  of



the minor axis; let  $x_1, y_1$  be the coordinates of the describing point  $P_1$ , and for the point  $Q$ , taking  $\phi$  as the complement of the eccentric angle, let the coordinates referred to the axes of the ellipse be

$$x_1 = a \sin \phi, \quad y_1 = b \cos \phi;$$

the arc  $BQ$  is then

$$= \int_0^\phi \sqrt{(a^2 \cos^2 \phi + b^2 \sin^2 \phi)} d\phi,$$

which is  $= aE\phi$ , the modulus being the eccentricity  $e$  of the ellipse; we then have for the coordinates of  $P_1$

$$\begin{aligned} x &= OP' = OQ - QP', & = BQ - QP', \\ y &= & P_1P', \end{aligned}$$

where  $QP'$  and  $P_1P'$  are the projections of  $P_1Q$  in the direction of, and at right angles to, the tangent  $QO$  at  $Q$ ; we find the values without difficulty, and we thus have

$$\begin{aligned} x &= \int_0^\phi \sqrt{(a^2 \cos^2 \phi + b^2 \sin^2 \phi)} d\phi - \frac{(a^2 - b^2) \sin \phi \cos \phi - ax_1 \cos \phi + by_1 \sin \phi}{\sqrt{(a^2 \cos^2 \phi + b^2 \sin^2 \phi)}}, \\ y &= \frac{ab - bx_1 \sin \phi - ay_1 \cos \phi}{\sqrt{(a^2 \cos^2 \phi + b^2 \sin^2 \phi)}}, \end{aligned}$$

which are the expressions for  $x, y$  in terms of a variable parameter  $\phi$ .

In particular, for the point  $B$  at the extremity of the minor axis,  $x_1 = 0$ ,  $y_1 = b$ , and the equations become

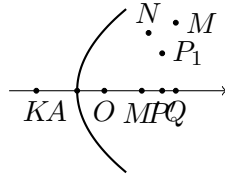
$$\begin{aligned} x &= \int_0^\phi \sqrt{(a^2 \cos^2 \phi + b^2 \sin^2 \phi)} d\phi - \frac{(a^2 - b^2) \sin \phi \cos \phi + b^2 \sin \phi}{\sqrt{(a^2 \cos^2 \phi + b^2 \sin^2 \phi)}}, \\ y &= \frac{ab(1 - \cos \phi)}{\sqrt{(a^2 \cos^2 \phi + b^2 \sin^2 \phi)}}, \end{aligned}$$

which, for  $a = b$ , become

$$x = a(\phi - \sin \phi), \quad y = a(1 - \cos \phi),$$

the ordinary formulæ for the cycloid.

In the case of the parabola  $Y^2 = 4aX$  (see figure), taking  $x_1, y_1$  for the coordinates



of the describing point  $P_1$ , and representing those of the point of contact  $Q$  by means of the parameter  $\phi$ ,  $X = a\phi^2$ ,  $Y = 2a\phi$ , then the arc  $AQ$  is

$$= a [\phi \sqrt{(\phi^2 + 1)} + \log\{\phi + \sqrt{(\phi^2 + 1)}\}],$$

so that, taking the origin at the point  $O$ , which comes into contact with the vertex  $A$ , this is also the value of  $OQ$ , and we then have

$$x = OQ - QP', \quad y = P_1P'.$$

Producing the axis of the parabola to meet the line  $y = 0$  in  $k$ , we have

$$AK = AN = a\phi^2,$$

and thence  $KM = a\phi^2 + x_1$ ; and then, if  $\theta$  be the inclination  $AKQ$  of the axis of the parabola,

$$KM' = (a\phi^2 + x_1) \cos \theta, \quad MM' = (a\phi^2 + x_1) \sin \theta;$$

and thence also

$$KP' = (a\phi^2 + x_1) \cos \theta + y_1 \sin \theta, \quad P_1P' = (a\phi^2 + x_1) \sin \theta - y_1 \cos \theta;$$

also

$$KQ = 2a\phi^2 \sec \theta,$$

and thence

$$QP' = 2a\phi^2 \sec \theta - (a\phi^2 + x_1) \cos \theta - y_1 \sin \theta.$$

We have  $\tan \theta = \frac{1}{\phi}$ , and thence

$$\sin \theta = \frac{1}{\sqrt{(\phi^2 + 1)}}, \quad \cos \theta = \frac{\phi}{\sqrt{(\phi^2 + 1)}}, \quad \sec \theta = \frac{\sqrt{(\phi^2 + 1)}}{\phi};$$

hence

$$QP' = 2a\phi\sqrt{(\phi^2 + 1)} - \frac{(a\phi^2 + x_1)\phi + y_1}{\sqrt{(\phi^2 + 1)}} = \frac{a(\phi^2 + 2\phi) - x_1\phi - y_1}{\sqrt{(\phi^2 + 1)}},$$

$$P_1P' = \frac{a\phi^2 + x_1 - \phi y_1}{\sqrt{(\phi^2 + 1)}},$$

and the values of  $x$ ,  $y$  thus become

$$x = a[\phi\sqrt{(\phi^2 + 1)} + \log\{\phi + \sqrt{(\phi^2 + 1)}\}] - \frac{a(\phi^2 + 2\phi) - x_1\phi - y_1}{\sqrt{(\phi^2 + 1)}},$$

that is,

$$x = a \log\{\phi + \sqrt{(\phi^2 + 1)}\} + \frac{(x_1 - a)\phi + y_1}{\sqrt{(\phi^2 + 1)}},$$

$$y = \frac{a\phi^2 + x_1 - \phi y_1}{\sqrt{(\phi^2 + 1)}}.$$

In particular, for the focus of the parabola,  $x_1 = a$ ,  $y_1 = 0$ , and the equations become

$$\begin{aligned}x &= a \log\{\phi + \sqrt{(\phi^2 + 1)}\}, \\y &= a \sqrt{(\phi^2 + 1)};\end{aligned}$$

that is,

$$\frac{x}{a} = \log\left\{\frac{y}{a} + \sqrt{\left(\frac{y^2}{a^2} - 1\right)}\right\},$$

or say

$$\exp\left(\frac{x}{a}\right) = \frac{y}{a} + \sqrt{\left(\frac{y^2}{a^2} - 1\right)};$$

this gives

$$\exp\left(-\frac{x}{a}\right) = \frac{y}{a} - \sqrt{\left(\frac{y^2}{a^2} - 1\right)};$$

and we have therefore

$$y = \frac{1}{2}a \left\{ \exp\left(\frac{x}{a}\right) + \exp\left(-\frac{x}{a}\right) \right\},$$

viz. as is well known, the epitrochoid described by the focus of the parabola is the catenary.

## 914.

## ON A SOLUBLE QUINTIC EQUATION.

[From the *American Journal of Mathematics*, vol. XIII. (1891), pp. 53–58.]

MR YOUNG, in his paper, “Solvable Quintic Equations with Commensurable Coefficients,” *American Journal of Mathematics*, x. (1888), pp. 99–130, has given, in illustration of his general theory of the solution of soluble quintic equations (founded upon a short note by Abel), no less than twenty instances of the solution of a quintic equation with purely numerical coefficients, having a solution of the form  $\sqrt[5]{A} + \sqrt[5]{B} + \sqrt[5]{C} + \sqrt[5]{D}$ , where  $A, B, C, D$  are numerical expressions involving only square roots. But the solutions are not presented in their most simple form: thus in example 1,  $x^5 + 3x^2 + 2x - 1 = 0$ , the expression involves a radical

$$\sqrt[5]{\frac{1}{4}(21125 + 9439\sqrt{5})} :$$

here

$$(21125+9439)\sqrt{5}, = \sqrt{5} (9439+4225\sqrt{5}), = \sqrt{5} \frac{1}{4}(18+5\sqrt{5})^2(1+\sqrt{5})^2(2+\sqrt{5}),$$

so that, taking out the roots of the squared factors, we have as the proper form of the radical the very much more simple form  $\sqrt[5]{47(2 + \sqrt{5})\sqrt{5}}$ ; where observe that  $(2 + \sqrt{5})(2 - \sqrt{5}) = -1$ , and thence  $(2 + \sqrt{5})\sqrt[5]{-47(2 - \sqrt{5})\sqrt{5}} = \sqrt[5]{47(2 + \sqrt{5})\sqrt{5}}$ , viz. the conjugate radicals  $\sqrt[5]{-47(2 - \sqrt{5})\sqrt{5}}$  and  $\sqrt[5]{47(2 + \sqrt{5})\sqrt{5}}$  differ only by a factor  $2 + \sqrt{5}$  which is rational in 1 and  $\sqrt{5}$ . To avoid fractions I consider the foregoing equation under the form

$$x^5 + 3000x^2 + 20000x - 100000 = 0,$$

and I will presently give the solution; but first I consider the general theory. Writing

$$\begin{aligned} A &= \alpha^5, & A' &= \alpha^2\gamma, & A'' &= \alpha^3\beta, \\ B &= \beta^5, & B' &= \alpha\beta^2, & B'' &= \beta^3\delta, \\ C &= \gamma^5, & C' &= \gamma^2\delta, & C'' &= \alpha\gamma^3, \\ D &= \delta^5, & D' &= \beta\delta^2, & D'' &= \gamma\delta^3, \end{aligned}$$



we have  $A'D' = \alpha^2\delta^2\beta\gamma$ ,  $B'C' = \alpha\delta\beta^2\gamma^2$ . Also

$$A'' = \frac{A'B'}{\beta\gamma}, \quad B'' = \frac{B'D'}{\alpha\delta}, \quad C'' = \frac{A'C'}{\alpha\delta}, \quad D'' = \frac{C'D'}{\beta\gamma},$$

which determine  $A'', B'', C'', D''$  in terms of  $A', B', C', D', \alpha\delta, \beta\gamma$ ; and then

$$A = \frac{A'A''}{\beta\gamma}, \quad B = \frac{B'B''}{\alpha\delta}, \quad C = \frac{C'C''}{\alpha\delta}, \quad D = \frac{D'D''}{\beta\gamma},$$

which give  $A, B, C, D$ .

If now we assume  $x = \alpha + \beta + \gamma + \delta$ , and regard  $A, B, C, D, A', B', C', D', A'', B'', C'', D'', \alpha\delta, \beta\gamma$  each as a rational function, we may express  $x, x^2, x^3, x^5$  each of them by means of rational functions or of rational functions multiplied into  $\alpha, \beta, \gamma, \delta$  respectively: thus,

$$\begin{aligned} x &= \alpha + \beta + \gamma + \delta, \\ x^2 &= \alpha^2 + \beta^2 + \gamma^2 + \delta^2 + 2\alpha\beta + 2\alpha\gamma + 2\alpha\delta + 2\beta\gamma + 2\beta\delta + 2\gamma\delta \\ &= \frac{A'\beta}{\beta\gamma} + \frac{B'\delta}{\alpha\delta} + \frac{C'\alpha}{\alpha\delta} + \frac{D'\gamma}{\beta\gamma} + \frac{2B'\gamma}{\beta\gamma} + \frac{2A'\delta}{\alpha\delta} + 2\alpha\delta + 2\beta\gamma + \frac{2D'\alpha}{\alpha\delta} + \frac{2C'\beta}{\beta\gamma}, \end{aligned}$$

&c.; and we thus obtain

$$\begin{aligned}
& x^5 + qx^3 + rx^2 + sx + t \\
= & A + B + C + D \\
& + (20\alpha\delta + 30\beta\gamma)(A' + D') + 30(\alpha\delta + 2\beta\gamma)(B' + C') \\
& + 3q(A' + B' + C' + D') + 2r(\alpha\delta + \beta\gamma) + t \\
& + \alpha \left\{ 5A'' + 5\frac{C'^2}{\alpha\delta} + \frac{5B'^2\beta\gamma}{\alpha\delta} + 10C'' + \frac{10D'^2}{\alpha\delta} + 10\alpha^2\delta^2 + 20D'' + 30\beta^2\gamma^2 + 30D''\frac{\beta\gamma}{\alpha\delta} + 60\alpha\beta\gamma\delta \right. \\
& \quad \left. + q\left(\frac{B'}{\alpha\delta} + \frac{3D'}{\alpha\delta} + 3\alpha\delta + 6\beta\gamma\right) + r\left(\frac{C'}{\alpha\delta} + \frac{2D'}{\alpha\delta}\right) + s \right\} \\
& + \beta \left\{ 5B'' + 5\frac{A'^2}{\beta\gamma} + 5\frac{D'^2\alpha\delta}{\beta\gamma} + 10A'' + \frac{10C''^2}{\beta\gamma} \right. \\
& \quad + 10\beta^2\gamma^2 + 20D'' + 20C'' + 30\alpha^2\delta^2 + 30C''\frac{\alpha\delta}{\beta\gamma} + 60\alpha\beta\gamma\delta \\
& \quad \left. + q\left(\frac{D'}{\beta\gamma} + \frac{3C'}{\beta\gamma} + 3\beta\gamma + 6\alpha\delta\right) + r\left(\frac{A'}{\beta\gamma} + \frac{2C'}{\beta\gamma}\right) + s \right\} \\
& + \gamma \left\{ 5C'' + 5\frac{D'^2}{\beta\gamma} + 5\frac{A''\alpha\delta}{\beta\gamma} + 10D'' + \frac{10B'^2}{\beta\gamma} \right. \\
& \quad + 10\beta^2\gamma^2 + 20A'' + 20B'' + 30\alpha^2\delta^2 + 30B''\frac{\alpha\delta}{\beta\gamma} + 60\alpha\beta\gamma\delta \\
& \quad \left. + q\left(\frac{A''}{\beta\gamma} + \frac{3B''}{\beta\gamma} + 3\beta\gamma + 6\alpha\delta\right) + r\left(\frac{D'}{\beta\gamma} + \frac{2B'}{\beta\gamma}\right) + s \right\} \\
& + \delta \left\{ 5D'' + 5\frac{B'^2}{\alpha\delta} + 5\frac{C''\beta\gamma}{\alpha\delta} + 10B'' + \frac{10A'^2}{\alpha\delta} \right. \\
& \quad + 10\alpha^2\delta^2 + 20C'' + 20A'' + 30\beta^2\gamma^2 + 30A''\frac{\beta\gamma}{\alpha\delta} + 60\alpha\beta\gamma\delta \\
& \quad \left. + q\left(\frac{C''}{\alpha\delta} + \frac{3A''}{\alpha\delta} + 3\alpha\delta + 6\beta\gamma\right) + r\left(\frac{B'}{\alpha\delta} + \frac{2A'}{\alpha\delta}\right) + s \right\}.
\end{aligned}$$

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If, then,  $x^5 + qx^3 + rx^2 + sx + t = 0$ , we have the rational term  $= 0$ , and the coefficients of  $\alpha, \beta, \gamma, \delta$  each  $= 0$ ; in the class of equations under consideration, these last equations differ only in the signs of the radicals contained therein, so that one of them being satisfied identically, the others will be also satisfied. In particular, if  $q = 0$ , then  $\alpha\delta + \beta\gamma = 0$ : the rational term gives

$$A + B + C + D - 10\alpha\delta(A' + D' - B' - C') + t = 0,$$

and the term in  $\alpha$  gives

$$5A'' + 15B'' + 10C'' - 10D'' + \frac{5}{\alpha\delta}(C'^2 + 2D'^2) + \frac{r}{\alpha\delta}(C' + 2D') - 20\alpha^2\delta^2 + s = 0.$$

For the equation  $x^5 + 3000x^2 + 20000x - 100000 = 0$ , the expression for the root is  $x = \sqrt[5]{A} + \sqrt[5]{B} + \sqrt[5]{C} + \sqrt[5]{D}$ , where

$$A = 39000 + 18200\sqrt{5} + (1720 + 920\sqrt{5})\sqrt{235 + 94\sqrt{5}},$$

$$D = 39000 + 18200\sqrt{5} + (-1720 - 920\sqrt{5})\sqrt{235 + 94\sqrt{5}},$$

$$B = 39000 - 18200\sqrt{5} + (-1720 + 920\sqrt{5})\sqrt{235 - 94\sqrt{5}},$$

$$C = 39000 - 18200\sqrt{5} + (1720 - 920\sqrt{5})\sqrt{235 - 94\sqrt{5}},$$

and where also

$$A' = -150 - 70\sqrt{5} + (-10 - 2\sqrt{5})\sqrt{235 + 94\sqrt{5}},$$

$$D' = -150 - 70\sqrt{5} + (10 + 2\sqrt{5})\sqrt{235 + 94\sqrt{5}},$$

$$B' = -150 + 70\sqrt{5} + (10 - 2\sqrt{5})\sqrt{235 - 94\sqrt{5}},$$

$$C' = -150 + 70\sqrt{5} + (-10 + 2\sqrt{5})\sqrt{235 - 94\sqrt{5}},$$

and

$$A'' = -940 - 100\sqrt{5} + (-100 + 20\sqrt{5})\sqrt{235 + 94\sqrt{5}},$$

$$D'' = -940 - 100\sqrt{5} + (100 - 20\sqrt{5})\sqrt{235 + 94\sqrt{5}},$$

$$B'' = -940 + 100\sqrt{5} + (100 + 20\sqrt{5})\sqrt{235 - 94\sqrt{5}},$$

$$C'' = -940 + 100\sqrt{5} + (-100 - 20\sqrt{5})\sqrt{235 - 94\sqrt{5}}.$$

The foregoing forms are in some respects the most convenient; but it is to be observed that we have

$$\begin{aligned} A &= 2600\sqrt{5}(1 + \sqrt{5})(2 + \sqrt{5}) + 40(1 + \sqrt{5})(18 + 5\sqrt{5})\sqrt{47\sqrt{5}(2 + \sqrt{5})}, \text{ \&c.}, \\ A' &= -10\sqrt{5}(1 + \sqrt{5})(2 + \sqrt{5}) - 2\sqrt{5}(1 + \sqrt{5})\sqrt{47\sqrt{5}(2 + \sqrt{5})}, \text{ \&c.}, \\ A'' &= 20(1 - \sqrt{5})(18 + 13\sqrt{5}) + 20\sqrt{5}(1 - \sqrt{5})\sqrt{47\sqrt{5}(2 + \sqrt{5})}, \text{ \&c.}, \end{aligned}$$

or putting for shortness

$$\sqrt{Q} = \sqrt{47\sqrt{5}(2 + \sqrt{5})}, \quad \sqrt{Q_1} = \sqrt{-47\sqrt{5}(2 - \sqrt{5})},$$

(so that, according to a foregoing remark, we have  $(2 + \sqrt{5})\sqrt{Q} = \sqrt{Q_1}$ ), then we have

$$\begin{aligned} A &= 40(1 + \sqrt{5})\{65\sqrt{5}(2 + \sqrt{5}) + (18 + 5\sqrt{5})\sqrt{Q}\}, \text{ \&c.}, \\ A' &= -2\sqrt{5}(1 + \sqrt{5})\{5(2 + \sqrt{5}) + \sqrt{Q}\}, \text{ \&c.}, \\ A'' &= 20(1 - \sqrt{5})\{18 + 13\sqrt{5} + \sqrt{5}\sqrt{Q}\}, \text{ \&c.}, \end{aligned}$$

where observe that the term  $2 + \sqrt{5}$  is a factor of  $Q$ .

Starting from the values of  $A'$ ,  $B'$ ,  $C'$ ,  $D'$ , we have

$$A' = -2\sqrt{5}(1 + \sqrt{5})\{5(2 + \sqrt{5}) + \sqrt{Q}\},$$

and therefore

$$D' = -2\sqrt{5}(1 + \sqrt{5})\{5(2 + \sqrt{5}) - \sqrt{Q}\},$$

$$A'D' = 20(1 + \sqrt{5})^2(2 + \sqrt{5})[25(2 + \sqrt{5}) - 47\sqrt{5}],$$

where the last factor is

$$= 50 - 22\sqrt{5}, = -2\sqrt{5}(11 - 5\sqrt{5}), = -\sqrt{5}(1 - \sqrt{5})^2(2 - \sqrt{5}).$$

Hence

$$A'D' = -\sqrt{5} \cdot 20(-4)^2(-1) = 320\sqrt{5},$$

that is,  $A'D' = (\alpha\delta)^2\beta\gamma = 320\sqrt{5}$ , and similarly  $B'C' = \alpha\delta(\beta\gamma)^2 = -320\sqrt{5}$ , whence  $\alpha\delta = -4\sqrt{5}$ ,  $\beta\gamma = 4\sqrt{5}$ , and  $\alpha\delta + \beta\gamma = 0$ , as above.

We have, moreover,

$$A' = -2\sqrt{5}(1 + \sqrt{5})\{5(2 + \sqrt{5}) + \sqrt{Q}\},$$

$$B' = 2\sqrt{5}(1 - \sqrt{5})\{5(2 - \sqrt{5}) - \sqrt{Q_1}\},$$

and thence

$$\begin{aligned} A'B' &= 80[-25 - \sqrt{Q}Q_1 + 5(2 - \sqrt{5})\sqrt{Q} - 5(2 + \sqrt{5})\sqrt{Q_1}] \\ &= 80[-25 - 47\sqrt{5} + (5(2 - \sqrt{5}) - 5)\sqrt{Q}], \end{aligned}$$

that is,

$$\begin{aligned} A'B' \div \beta\gamma &= 4\sqrt{5}[-25 - 47\sqrt{5} + 5(1 - \sqrt{5})\sqrt{Q}] \\ &= -20(47 + 5\sqrt{5}) + 20\sqrt{5}(1 - \sqrt{5})\sqrt{Q}, = A''; \end{aligned}$$

and similarly we verify the values of  $B''$ ,  $C''$  and  $D''$ .

We have next

$$A'A'' = 160\sqrt{5} \{ (10 + 5\sqrt{5} + \sqrt{Q})(18 + 13\sqrt{5} + \sqrt{5}\sqrt{Q}) \},$$

or observing that  $Q\sqrt{5}$  is  $= 235(2 + \sqrt{5})$ , the whole term in  $\{ \}$  is

$$= (505 + 220\sqrt{5}) + (470 + 235\sqrt{5}) + (18 + 13\sqrt{5} + 25 + 10\sqrt{5})\sqrt{Q},$$

or we have

$$\begin{aligned} A'A'' &= 160\sqrt{5} \{ 65\sqrt{5} (7 + 3\sqrt{5}) + (43 + 23\sqrt{5})\sqrt{Q} \} \\ &= 160\sqrt{5} (1 + \sqrt{5}) \{ 65\sqrt{5} (2 + \sqrt{5}) + (18 + 5\sqrt{5})\sqrt{Q} \}, \end{aligned}$$

and consequently

$$A'A'' \div \beta\gamma = 40(1 + \sqrt{5}) \{ 65\sqrt{5} (2 + \sqrt{5}) + (18 + 5\sqrt{5})\sqrt{Q} \}, = A;$$

and similarly we verify the values of  $B$ ,  $C$ ,  $D$ .

12-2

In the proposed equation  $x^5 + 3000x^2 + 20000x - 100000 = 0$ , we have  $r = 3000$ ,  $s = 20000$ ,  $t = -100000$ ,  $\alpha\delta = -4\sqrt{5}$ ; the two equations to be verified thus are

$$A + B + C + D + 40\sqrt{5}(A' + D' - B' - C') - 100000 = 0,$$

and

$$5A'' + 15B'' + 10C'' - 10D'' - \frac{\sqrt{5}}{4}(C'^2 + 2D'^2) - 150\sqrt{5}(C' + 2D') - 1600 + 20000 = 0.$$

As to the first of these, we have  $A + B + C + D = 156000$ ,  $A' + D' - B' - C' = -280\sqrt{5}$ , and the equation thus is

$$156000 + 40\sqrt{5}(-280\sqrt{5}) - 100000 = 0,$$

which is right.

For the second equation, if in the calculation we keep the radicals in the first instance distinct, we have

$$\begin{aligned} 5A'' + 15B'' + 10C'' - 10D'' &= -18800 + 3000\sqrt{5} + (-1500 + 300\sqrt{5})\sqrt{Q} + (500 + 100\sqrt{5})\sqrt{Q_1}, \\ -150\sqrt{5}(C' + 2D') &= \{-450 - 70\sqrt{5} \\ &\quad + (20 + 4 + \sqrt{5})\sqrt{Q} + (-10 + 2\sqrt{5})\sqrt{Q_1}\}(-150\sqrt{5}) \\ -1600 + 20000 &= 18400, \\ -\frac{\sqrt{5}}{4}(C'^2 + 2D'^2) &= -\frac{\sqrt{5}}{4}\{282000 + 416800\sqrt{5} \\ &\quad + (-8800 - 4000\sqrt{5})\sqrt{Q} + (4400 - 2000\sqrt{5})\sqrt{Q_1}\}. \end{aligned}$$

Substituting in the equation, we ought to have

$$\begin{aligned} 0 &= -18800 + 3000\sqrt{5} + (-1500 + 300\sqrt{5})\sqrt{Q} + (500 + 100\sqrt{5})\sqrt{Q_1} \\ &\quad + 52500 + 67500\sqrt{5} + (-3000 - 3000\sqrt{5})\sqrt{Q} + (-1500 + 1500\sqrt{5})\sqrt{Q_1} \\ &\quad + 18400 \\ &\quad - 52100 - 70500\sqrt{5} + (5000 + 2200\sqrt{5})\sqrt{Q} + (2500 - 1100\sqrt{5})\sqrt{Q_1}, \end{aligned}$$

that is,

$$0 = (500 - 500\sqrt{5})\sqrt{Q} + (1500 + 500\sqrt{5})\sqrt{Q_1},$$

viz.

$$0 = 500\{(1 - \sqrt{5})\sqrt{Q} + (3 + \sqrt{5})\sqrt{Q_1}\},$$

which is satisfied in virtue of  $\sqrt{Q} = (2 + \sqrt{5})\sqrt{Q_1}$ : this completes the verification.



## 915.

## ON THE PARTITIONS OF A POLYGON.

[From the *Proceedings of the London Mathematical Society*, vol. XXII. (1891), pp. 237–262. Read March 12, 1891.]

1. THE partitions are made by non-intersecting diagonals; the problems which have been successively considered are (1) to find the number of partitions of an  $r$ -gon into triangles, (2) to find the number of partitions of an  $r$ -gon into  $k$  parts, and (3) to find the number of partitions of an  $r$ -gon into  $p$ -gons,  $r$  of the form

$$n(p-2)+2.$$

Problem (1) is a particular case of (2); and it is also a particular case of (3); but the problems (2) and (3) are outside each other; for problem (3) a very elegant solution, which will be here reproduced, is given in the paper, H. M. Taylor and R. C. Rowe, “Note on a Geometrical Theorem,” *Proc. Lond. Math. Soc.*, t. XIII. (1882), pp. 102–106, and this same paper gives the history of the solution of (1).

The solution of (2) is given in the memoir, Kirkman “On the  $k$ -partitions of the  $r$ -gon and  $r$ -ace,” *Phil. Trans.*, t. CXLVII. (for 1857), p. 225; viz. he there gives for the number of partitions of the  $r$ -gon into  $k$  parts (or, what is the same thing, by means of  $k-1$  non-intersecting diagonals) the expression

$$\frac{[r+k-2]^{k-1} [r-3]^{k-1}}{[k]^{k-1} [k-1]^{k-1}},$$

but there is no complete demonstration of this result.

If  $k = r - 2$ , we have the solution of the problem (1); viz. the number of partitions of the  $r$ -gon into triangles is

$$= [2r-4]^{r-3} \div [r-2]^{r-3}.$$

The present paper relates chiefly to the foregoing problem (2), the determination

of the number of partitions of the  $r$ -gon into  $k$  parts, or, what is the same thing, by means of  $k - 1$  non-intersecting diagonals.

2. Assuming for the moment the foregoing result, then for  $k = 1$ , the number of partitions is

$$= 1,$$

for  $k = 2$  it is

$$= \frac{r.r - 3}{2},$$

for  $k = 3$  it is

$$= \frac{r + 1.r.r - 3.r - 4}{12},$$

and so on. As a simple verification,  $k = 2$ , the number of partitions is equal to the number of diagonals, viz. this is number of pairs of summits less number of sides, that is,

$$\frac{1}{2}r(r - 1) - r, = \frac{1}{2}r(r - 3).$$

For convenience, I give the first Table on the next page, which is a tabulation of the functions

$$\begin{aligned} U_1 &= x^3 + x^4 + x^5 + x^6 + \dots + x^r, \\ U_2 &= 2x^4 + 5x^5 + 9x^6 + \dots + \frac{r.r - 3}{2.1} x^r, \\ U_3 &= 5x^5 + 21x^6 + \dots + \frac{r + 1.r.r - 3.r - 4}{3.2.2.1} x^r, \\ U_4 &= 14x^6 + \dots + \frac{r + 2.r + 1.r.r - 3.r - 4.r - 5}{4.3.2.3.2.1} x^r, \\ U_5 &= \dots + \frac{r + 3.r + 2.r + 1.r.r - 3.r - 4.r - 5.r - 6}{5.4.3.2.4.3.2.1} x^r, \\ &\&c. \end{aligned}$$

3. And in connexion herewith, I give the second Table on the next page, which is a tabulation of the functions

$$\begin{aligned} V_2 &= x^4 + 2x^5 + 3x^6 + 4x^7 + \dots + \frac{r-1}{1} x^{r+2}, \\ V_3 &= 4x^6 + 14x^7 + 32x^8 + \dots + 2 \frac{r+1.r-3.r-4}{3.2.1} x^{r+2}, \\ V_4 &= 14x^8 + 72x^9 + \dots + 3 \frac{r+2.r+1.r-3.r-4.r-5}{4.3.3.2.1} x^{r+2}, \\ V_5 &= 48x^{10} + \dots + 4 \frac{r+3.r+2.r+1.r-3.r-4.r-5.r-6}{5.4.3.3.2.1} x^{r+2}, \\ &\&c. \end{aligned}$$

| $r =$ | $k=1$ | 2  | 3    | 4     | 5      | 6       | 7       | 8        | 9        | 10       | 11       | 12      | 13     |
|-------|-------|----|------|-------|--------|---------|---------|----------|----------|----------|----------|---------|--------|
| 3     | 1     |    |      |       |        |         |         |          |          |          |          |         |        |
| 4     | 1     | 2  |      |       |        |         |         |          |          |          |          |         |        |
| 5     | 1     | 5  | 5    |       |        |         |         |          |          |          |          |         |        |
| 6     | 1     | 9  | 21   | 14    |        |         |         |          |          |          |          |         |        |
| 7     | 1     | 14 | 56   | 84    | 42     |         |         |          |          |          |          |         |        |
| 8     | 1     | 20 | 120  | 300   | 330    | 132     |         |          |          |          |          |         |        |
| 9     | 1     | 27 | 225  | 825   | 1485   | 1287    | 429     |          |          |          |          |         |        |
| 10    | 1     | 35 | 385  | 1925  | 5005   | 7007    | 5005    | 1430     |          |          |          |         |        |
| 11    | 1     | 44 | 616  | 4004  | 14014  | 28028   | 32032   | 19448    | 4862     |          |          |         |        |
| 12    | 1     | 54 | 936  | 7644  | 34398  | 91728   | 148512  | 143208   | 75582    | 16796    |          |         |        |
| 13    | 1     | 65 | 1365 | 13650 | 76440  | 259896  | 556920  | 755820   | 629850   | 293930   | 58786    |         |        |
| 14    | 1     | 77 | 1925 | 23100 | 157080 | 659736  | 1790712 | 3197700  | 3730650  | 2735810  | 1144066  | 208012  |        |
| 15    | 1     | 90 | 2640 | 37400 | 302940 | 1534896 | 5116320 | 11511720 | 17587350 | 17978180 | 11767536 | 4457400 | 742900 |

| $r =$ | $k=2$ | 3   | 4     | 5      | 6       | 7       | 8        | 9        | 10       | 11       | 12      | 13      |
|-------|-------|-----|-------|--------|---------|---------|----------|----------|----------|----------|---------|---------|
| 4     | 1     |     |       |        |         |         |          |          |          |          |         |         |
| 5     | 2     | 4   |       |        |         |         |          |          |          |          |         |         |
| 6     | 3     | 14  | 14    |        |         |         |          |          |          |          |         |         |
| 7     | 4     | 32  | 72    | 48     |         |         |          |          |          |          |         |         |
| 8     | 5     | 60  | 225   | 330    | 165     |         |          |          |          |          |         |         |
| 9     | 6     | 100 | 550   | 1320   | 1430    | 572     |          |          |          |          |         |         |
| 10    | 7     | 154 | 1155  | 4004   | 7007    | 6006    | 2002     |          |          |          |         |         |
| 11    | 8     | 224 | 2184  | 10192  | 25480   | 34944   | 24052    | 7072     |          |          |         |         |
| 12    | 9     | 312 | 3822  | 22932  | 76440   | 148512  | 167076   | 100776   | 25194    |          |         |         |
| 13    | 10    | 420 | 6300  | 47040  | 199920  | 514080  | 813960   | 775200   | 396800   | 90440    |         |         |
| 14    | 11    | 550 | 9900  | 89760  | 476240  | 1534996 | 3197700  | 4263600  | 3517470  | 1634880  | 326876  |         |
| 15    | 12    | 704 | 14960 | 161568 | 1023264 | 4093056 | 10744272 | 18759840 | 18573816 | 15690048 | 6547520 | 1188640 |

4. These functions,  $U$  and  $V$ , are particular values satisfying the equation

$$(V_2 + V_3y + V_4y^2 + \dots) = (U_1 + U_2y + U_3y^2 + U_4y^3 + \dots)^2;$$

that this is so will appear from the following general investigation.

5. Taking  $x, y$  as independent variables, denoting by  $X$  an arbitrary function of  $x$ , and using accents to denote differentiations in regard to  $x$ , we require the following identity:

$$\frac{2}{1.2}X^2 + \frac{4y}{1.2.3}(X^2)' + \frac{6y^2}{1.2.3.4}(X^4)'' + \dots = \left\{ X + \frac{y}{1.2}(X^2)' + \frac{y^2}{1.2.3}(X^3)'' + \dots \right\}^2,$$

which I prove as follows. Writing  $U$  to denote the same function of  $u$  which  $X$  is of  $x$ , I start from the equation

$$u = x + yU,$$

which determines  $u$  as a function of the independent variables  $x, y$ . We have

$$\frac{du}{dy}(1 - yU') = U, \quad \frac{du}{dx}(1 - yU') = 1,$$

where the accent denotes differentiation in regard to  $u$ ; hence

$$\frac{du}{dy} = U \frac{du}{dx} = \frac{u - x}{y} \frac{du}{dx},$$

or say

$$y \frac{du}{dy} = (u - x) \frac{du}{dx}.$$

Writing

$$u_1 = \int u \, dx,$$

and therefore

$$\frac{du_1}{dx} = u,$$

this equation may be written

$$y \frac{d^2 u_1}{dx \, dy} = \frac{du_1}{dx} - x - y \frac{du}{dx};$$

or, integrating with respect to  $x$ , we have

$$y \frac{du_1}{dy} = u_1 - \frac{1}{2}x^2 - yu,$$

or say

$$\frac{2}{y} \frac{d}{dy} \left( \frac{u_1 - \frac{1}{2}x^2}{y} \right) = \frac{(u - x)^2}{y^2},$$

that is,

$$2 \frac{d}{dy} \left( \frac{u_1 - \frac{1}{2}x^2}{y} \right) = \frac{(u - x)^2}{y^2}.$$

6. But, from the equation  $u = x + yU$ , we have

$$u = x + yX + \frac{y^2}{1.2}(X^2)' + \frac{6y^3}{1.2.3.4}(X^4)'' + \dots,$$

and thence

$$u_1 = \frac{1}{2}x^2 + yX_1 + \frac{y^2}{1.2}X^2 + \frac{y^3}{1.2.3}(X^3)' + \frac{y^4}{1.2.3.4}(X^4)'' + \dots,$$

if for a moment  $X_1$  is written for  $\int X dx$ . And hence, from the relation obtained above, we have the required identity

$$\frac{2}{1.2}X^2 + \frac{4y}{1.2.3}(X^2)' + \frac{6y^2}{1.2.3.4}(X^4)'' + \dots = \left\{ X + \frac{y}{1.2}(X^2)' + \frac{y^2}{1.2.3}(X^3)'' + \dots \right\}^2.$$

This of course gives the series of identities

$$\frac{2}{1.2}X^2 = X^2,$$

$$\frac{4}{1.2.3}(X^2)' = \frac{2}{1.2}X(X^2)',$$

$$\frac{6}{1.2.3.4}(X^4)'' = \frac{2}{1.2.3}X(X^3)'' + \left\{ \frac{1}{1.2}(X^2)' \right\}^2,$$

⋮

or say

$$X^2 = X^2,$$

$$(X^2)' = \frac{1}{2}X(X^2)',$$

$$(X^4)'' = \frac{1}{2}X(X^3)'' + [(X^2)']^2,$$

⋮

all of which may be easily verified.

7. I multiply each side of the identity by  $x^2$ , and write

$$\begin{aligned} U_1 &= x.X, & V_2 &= x^2 \cdot \frac{2}{1.2}X^2, \\ U_2 &= x \cdot \frac{1}{1.2}(X^2)', & V_3 &= x^2 \cdot \frac{\frac{1}{4}}{1.2.3}(X^2)', \\ U_3 &= x \cdot \frac{1}{1.2.3}(X^3)'', & V_4 &= x^2 \cdot \frac{6}{1.2.3.4}(X^4)'', \\ U_4 &= x \cdot \frac{1}{1.2.3.4}(X^4)''', & V_5 &= x^2 \cdot \frac{8}{1.2.3.4.5}(X^5)''', \\ & \vdots & & \end{aligned}$$

We thus obtain two sets of functions  $U$  and  $V$ , satisfying the before-mentioned equation. We have

$$(V_2 + yV_3 + y^2V_4 + \dots) = (U_1 + yU_2 + y^2U_3 + \dots)^2;$$

and it will be observed that we have, moreover, the relations

$$U_2 = \frac{1}{2}x(x^{-2}V_2)', \quad U_3 = \frac{1}{2}x(x^{-2}V_3)', \quad U_4 = \frac{1}{2}x(x^{-2}V_4)', \quad \dots$$

8. In particular, if

$$X = \frac{x^2}{1-x},$$

then the general term

$$\begin{array}{ll}
 \text{in } X \text{ is } x^{r-1}, & \text{the first term occurring when } r = 3, \\
 \text{in } X^2 \text{ is } (r-3)x^r, & \text{,, ,, } r = 4, \\
 \text{in } X^3 \text{ is } \frac{r-3 \cdot r-4}{1 \cdot 2} x^{r+1}, & \text{,, ,, } r = 5, \\
 \text{in } X^4 \text{ is } \frac{r-3 \cdot r-4 \cdot r-5}{1 \cdot 2 \cdot 3} x^{r+2}, & \text{,, ,, } r = 6,
 \end{array}$$

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from which it appears that, for this value of  $X$ ,  $U_1$ ,  $U_2$ ,  $U_3$ ,  $U_4$ , &c., have the before-mentioned values (No. 2), and further that  $V_2$ ,  $V_3$ ,  $V_4$ ,  $V_5$ , &c., have also the before-mentioned values (No. 3).

9. We do not absolutely require, but it is interesting to obtain, the finite expressions of these functions. We have

$$\begin{aligned}
 (1-x)U_1 &= x^3(1), \\
 (1-x)^3U_2 &= x^4(2-x), \\
 (1-x)^5U_3 &= x^5(5-4x+x^2), \\
 (1-x)^7U_4 &= x^6(14-14x+6x^2-x^3), \\
 (1-x)^9U_5 &= x^7(42-48x+27x^2-8x^3+x^4), \\
 (1-x)^{11}U_6 &= x^8(132-165x+110x^2-44x^3+10x^4-x^5), \\
 &\vdots \\
 (1-x)^2V_2 &= x^4(1), \\
 (1-x)^4V_3 &= x^6(4-2x), \\
 (1-x)^6V_4 &= x^8(14-12x+3x^2), \\
 (1-x)^8V_5 &= x^{10}(48-54x+24x^2-4x^3), \\
 &\vdots
 \end{aligned}$$

and here the factors in ( ) satisfy the series of relations

$$\begin{aligned}
 1 &= 1^2, \\
 4-2x &= 2(2-x), \\
 14-12x+3x^2 &= 2 \cdot 1 \cdot (5-4x+x^2) + (2-x)^2, \\
 48-54x+24x^2-4x^3 &= 2 \cdot 1 \cdot (14-14x+6x^2-x^3) + 2(2-x)(5-4x+x^2), \\
 &\vdots
 \end{aligned}$$

corresponding to

$$V_2 = U_1^2, \quad V_3 = 2U_1U_2, \quad \&c.,$$

given by the before-mentioned equation (No. 7), between the functions  $V$  and  $U$ .

10. It is to be shown that, taking  $U_1$ ,  $U_2$ ,  $U_3$ , ... for the functions which belong to the partitions of the  $r$ -gon (assumed to be unknown functions of



$r$  and the suffixes), and connecting them with a set of functions  $V_2, V_3, V_4, \dots$  by the relations

$$U_2 = \frac{1}{2}x(x^{-2}V_2)', \quad U_3 = \frac{1}{2}x(x^{-2}V_3)', \quad U_4 = \frac{1}{2}x(x^{-2}V_4)', \quad \&c.,$$

then we have the foregoing identical equation

$$(V_2 + V_3y + V_4y^2 + \dots) = (U_1 + U_2y + U_3y^2 + U_4y^3 + \dots)^2.$$

This implies the relations

$$\begin{aligned} V_2 &= U_1^2, \\ V_3 &= 2U_1U_2, \\ V_4 &= 2U_1U_3 + U_2^2, \\ V_5 &= 2U_1U_4 + 2U_2U_3, \\ &\text{\&c.} \end{aligned}$$

Thus, if  $U_1$  is known, the equation

$$V_2 = U_1^2$$

determines  $V_2$ , and then the equation

$$U_2 = \frac{1}{2}x(x^{-2}V_2)'$$

determines  $U_2$ , so that  $U_1, U_2$  are known; and we thence in the same way find successively  $U_3$  and  $V_3$ ,  $U_4$  and  $V_4$ , and so on; that is, assuming only that  $U_1$  has the before-mentioned value,

$$U_1 = x^3 + x^4 + x^5 + \dots + x^r + \dots,$$

it follows that all the remaining functions  $U$  and  $V$  must have their before-mentioned values. But the function

$$U_1 = x^3 + x^4 + x^5 + \dots,$$

where each coefficient is  $= 1$ , is evidently the proper expression for the generating function of the number of partitions of the  $r$ -gon into a single part; and we thus arrive at the proof that the remaining functions  $U$ , which are the generating functions for the number of partitions of the  $r$ -gon into 2, 3, 4,  $\dots$ ,  $k$ , parts, have their before-mentioned values.

11. Considering, then, the partition problem from the point of view just referred to, I write  $A_r, B_r, C_r, \dots$  for the number of partitions of an  $r$ -gon into 1 part, 2 parts, 3 parts, &c., and form therewith the generating functions

$$\begin{aligned} U_1 &= A_3x^3 + A_4x^4 + \dots + A_rx^r + \dots, \\ U_2 &= B_4x^4 + \dots + B_rx^r + \dots, \\ U_3 &= C_5x^5 + \dots + C_rx^r + \dots, \\ &\vdots \end{aligned}$$

and also the functions

$$\begin{aligned} V_2 &= \frac{3}{2}B_4x^6 + \dots + \frac{r-1}{r}B_rx^{r+2} + \dots, \\ V_3 &= \frac{4}{3}C_5x^7 + \dots + \frac{4}{r}C_rx^{r+2} + \dots, \\ &\vdots \end{aligned}$$

where observe that the functions  $U, V$  are such that

$$U_2 = \frac{1}{2}x(x^{-2}V_2)', \quad U_3 = \frac{1}{2}x(x^{-2}V_3)', \quad U_4 = \frac{1}{2}x(x^{-2}V_4)', \quad \&c.$$

To fix the ideas, consider an  $r$ -gon which is to be divided into six parts. Choosing any particular summit, and from this summit drawing a diagonal successively

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to each of the non-adjacent  $r - 3$  summits, we divide the  $r$ -gon into two parts in  $r - 3$  different ways; viz. the two parts are

$$\begin{aligned} & \text{a 3-gon and } (r - 1)\text{-gon,} \\ & \text{4-gon and } (r - 2)\text{-gon,} \\ & \vdots \\ & (r - 1)\text{-gon and 3-gon;} \end{aligned}$$

say any one of these ways is

$$\text{an } a\text{-gon and } \beta\text{-gon, } \alpha + \beta = r + 2.$$

Next, writing

$$a + b = 6,$$

that is,

$$a, b = 1, 5, \quad 2, 4, \quad 3, 3, \quad 4, 2, \quad 5, 1,$$

we divide in every possible way the  $a$ -gon into  $a$  parts, and the  $\beta$ -gon into  $b$  parts (so dividing the  $r$ -gon into six parts). Observing that  $A, B, C, D, E, F$  are the letters belonging to the numbers 1, 2, 3, 4, 5, 6, respectively, the number of parts which we thus obtain (corresponding to the different values of  $a, b$ ) are

$$A_\alpha E_\beta + B_\alpha D_\beta + C_\alpha C_\beta + D_\alpha B_\beta + E_\alpha A_\beta,$$

and summing for the different values of  $\alpha, \beta$  ( $\alpha + \beta = r + 2$ ), the whole number of parts is

$$= \text{coeff. } x^{r+2} \text{ in } (U_1 U_5 + U_2 U_4 + U_3 U_3 + U_4 U_2 + U_5 U_1),$$

that is,

$$= \text{coeff. } x^{r+2} \text{ in } (2U_1 U_5 + 2U_2 U_4 + U_3^2).$$

12. To obtain the whole number of the partitions of the  $r$ -gon into six parts, we must perform the foregoing process successively with each summit of the  $r$ -gon as the summit from which is drawn the diagonal which divides the  $r$ -gon into two parts; that is, the number found as above is to be multiplied by  $r$ . We thus obtain all the partitions repeated a certain number of times, viz. each partition into six parts is a partition by means of five diagonals, and is thus obtainable by the foregoing process, taking any one of the ten extremities of these diagonals as the point from which is

drawn the diagonal which divides the  $r$ -gon into two parts; that is, we have to divide the foregoing product by 10. The final result thus is

$$\frac{r}{10} F_r = \text{coeff. } x^{r+2} \text{ in } (2U_1U_5 + 2U_2U_4 + U_3^2),$$

where

$$\frac{r}{10} F_r \text{ is } = \text{coeff. } x^{r+2} \text{ in } V_6;$$

we thus have

$$V_6 = 2U_1U_5 + 2U_2U_4 + U_3^2.$$

13. The reasoning is perfectly general; and applying it successively to the partitions into two parts, three parts, &c., we have

$$\begin{aligned} V_2 &= U_1^2, \\ V_3 &= 2U_1U_2, \\ V_4 &= 2U_1U_3 + U_2^2, \\ V_5 &= 2U_1U_4 + 2U_2U_3, \\ &\vdots \end{aligned}$$

where any function  $V$  is related to the corresponding function  $U$  as above. The value of  $U_1$  is obviously

$$U_1 = x^3 + x^4 + x^5 + \dots = \frac{x^3}{1-x};$$

and hence the several functions  $U$  and  $V$  have the values above written down; the general term of  $U_k$  is

$$\frac{[r+k-2]^{k-1} [r-3]^{k-1}}{[k]^{k-1} [k-1]^{k-1}} x^r,$$

and the number of partitions of the  $r$ -gon into  $k$  parts is equal to the coefficient of  $x^r$  in this general term.

14. In the investigations which next follow, I consider, without using the method of generating functions, the problem of the partition of the  $r$ -gon into 2, 3, 4 or 5 parts; it will be convenient to state the results as follows:

Number of Partitions.

$$\begin{aligned} 2 \text{ parts, } & \frac{r}{2} A, \\ 3 \text{ parts, } & \frac{r}{4} 2A, \\ 4 \text{ parts, } & \frac{r}{6} (3A + 2B), \\ 5 \text{ parts, } & \frac{r}{8} (4A + 8B + 2C); \end{aligned}$$

where the capital letters refer to different "diagonal-types," thus:

| 2 parts,<br>or 1 diagonal: | 3 parts,<br>or 2 diagonals: | 4 parts,<br>or 3 diagonals: | 5 parts,<br>or 4 diagonals: |
|----------------------------|-----------------------------|-----------------------------|-----------------------------|
| $A$                        | $A$                         | $A \quad B$                 | $A \quad B \quad C$         |
|                            |                             | $\triangle$                 | $<  \quad \square$          |

viz. if, in a polygon divided into  $k$  parts by means of  $k - 1$  diagonals, we delete all the sides of the polygon, leaving only the diagonals, then these will present themselves under distinct forms, which are what I call “diagonal-types”; for instance, when

$$k = 4,$$

there are the two types  $A$  and  $B$  shown in the above diagram for four parts.

15. It is to be observed that we have sub-types corresponding to the coalescence of the terminal points of different diagonals; thus, suppose

$$k = 4.$$

Writing now  $A^\circ$  and  $B^\circ$  to denote the forms without coalescences, we have the sub-types  $A^\circ$ ,  $A'$ ,  $A''$  and  $B^\circ$ ,  $B'$ ,  $B''$ ,  $B'''$ , as follows:

$$\begin{array}{ccccccc} & \text{4 parts, or 3 diagonals:} \\ A^\circ & A' & A'' & B^\circ & B' & B'' & B''' \\ ||| & \wedge | & \swarrow \wedge & \triangle & \triangle & \triangle & \triangle \end{array}$$

where observe that under  $A''$  are included two distinct forms, which, nevertheless, by reason that there is in each of them the same number ( $= 2$ ) of coalescences, are reckoned as belonging to the same sub-type.

16. The numbers called  $A$ ,  $B$ ,  $C$ , &c., have values which may be directly determined. I write down as follows:

$$\text{1 diagonal, } A = \frac{r-3}{1}.$$

$$\text{2 diagonals, } A = \frac{r-3.r-2.r-1}{6} - 2 \frac{r-3}{1} = \frac{r-3.r-4}{6}(r+1);$$

where the calculation is

$$\begin{array}{r} r-2.r-1 = r^2-3r+2 \\ -6 \quad -6 \\ \hline = r^2-3r-4 \\ = r-4 \cdot r+1. \end{array}$$

$$\begin{aligned} \text{3 diagonals, } A &= \frac{r-3.r-2.r-1.r.r+1}{120} - 2 \frac{r-3.r-2.r-1}{6} + 1 \cdot \frac{r-3}{1} \\ &= \frac{r-3.r-2.r-1.r.r+1}{120} - 2 \frac{r-3.r-2.r-1}{6} + 1 \cdot \frac{r-3}{1} \\ &= \frac{r-3.r-4.r-5}{120}(r^2+7r+2); \end{aligned}$$



where the calculation is

$$\begin{array}{r}
 r - 2r - 1 \cdot r \cdot r + 1 = r^4 - 2r^3 - r^2 + 2r \\
 -40 \cdot r - 2r - 1 \quad -40r^2 + 120r - 80 \\
 +1240 \quad +1240 \\
 \hline
 = r^4 - 2r^3 - 41r^2 + 122r + 40 \\
 = r - 4 \cdot r - 5 \cdot r^2 + \dots \\
 \hline
 \frac{r - 5r - 4r - 3r - 2r - 1}{120}.
 \end{array}$$

$$\begin{aligned}
4 \text{ diagonals, } A &= \frac{r - 3.r - 2.r - 1.r.r + 1.r + 2.r + 3}{5040} \\
&- 3 \frac{r - 3.r - 2.r - 1.r.r + 1}{120} + 3 \frac{r - 3.r - 2.r - 1}{6} - 1 \frac{r - 3}{1} \\
&= \frac{r - 3.r - 4.r - 5.r - 6}{5040} (r^3 + 18r^2 + 65r);
\end{aligned}$$

where the calculation is

$$\begin{aligned}
r - 2.r - 1.r.r + 1.r + 2.r + 3 &= r^6 + 3r^5 - 5r^4 - 15r^3 + 4r^2 + 12r \\
&- 126 \cdot r - 2.r - 1.r.r + 1 & - 126r^4 + 252r^3 + 126r^2 - 252r \\
&+ 2520 \cdot r - 2.r - 1 & + 2520r^2 - 7560r + 5040 \\
&- 5040 & - 5040 \\
&\hline
&= 2650r^2 - 7800r \\
&= r - 4 \cdot r - 5 \cdot r - 6 \cdot r^3 + 18r^2 + 65r.
\end{aligned}$$

Also

$$\begin{aligned}
B &= \frac{r - 5.r - 4.r - 3.r - 2.r - 1.r.r + 1}{5040} - \frac{r - 5.r - 4.r - 3.r - 2.r - 1}{120} \\
&= \frac{r - 5.r - 4.r - 3.r - 2.r - 1.r.r + 1}{5040} - \frac{r - 5.r - 4.r - 3.r - 2.r - 1 \cdot 42}{5040} \\
&= \frac{r - 3.r - 4.r - 5.r - 6}{5040} (r - 1.r - 2.r + 7), \\
C &= \frac{r - 7.r - 6.r - 5.r - 4.r - 3.r - 2.r - 1}{5040} \\
&= \frac{r - 3.r - 4.r - 5.r - 6}{5040} (r - 1.r - 2.r - 7);
\end{aligned}$$

where the calculation is

$$\begin{aligned}
r.r + 1 &= r^2 + r \\
&- 42 & - 42 \\
&\hline
&= r^2 + r - 42 \\
&= r - 6 \cdot r + 7.
\end{aligned}$$

17. To explain the formation of these expressions, observe that:

*One diagonal.*—There must be on each side of the diagonal, or say in each of the two “intervals” formed by the diagonal, two sides; there remain  $r - 4$  sides which may be distributed at pleasure between the two intervals, and the number of ways in which this can be done is

$$= \frac{r-3}{1}.$$

*Two diagonals.*—There must be on each side of the two diagonals, formed by the two diagonals, two sides; there remain  $r - 4$  sides to be distributed between the same four intervals, and the number of ways in which this can be done is

$$\frac{r-3.r-2.r-1}{6}.$$

But we must exclude the distributions where there is no side in the one interval and no side in the other interval between the two diagonals; the number of these is that for the case of the coalescence of the two diagonals into a single diagonal, viz. it is

$$= \frac{r-3}{1};$$

and thus the number required is

$$\frac{r-3.r-2.r-1}{6} - 2 \frac{r-3}{1}.$$

18. *Three diagonals, A.*—There must be on each side of the three diagonals, that is, in two of the six intervals formed by the diagonals, two sides; there remain  $r-4$  sides to be distributed between the same six intervals, and the number of ways in which this can be done is

$$= \frac{r-3.r-2.r-1.r+1}{120}.$$

But we must exclude distributions which would permit the coalescence of the first and second, or of the second and third, or of all three of the diagonals. For the coalescence of the first and second diagonals (the third diagonal not coalescing), the term to be subtracted is

$$\frac{r-3.r-2.r-1}{6} - \frac{r-3}{1},$$

and the same number for the coalescence of the second and third diagonals (the first diagonal not coalescing); that is, the last-mentioned number is to be multiplied by 2; and for the coalescence of all three diagonals the number to be subtracted is

$$= \frac{r-3}{1};$$

we have thus the foregoing value

$$\frac{r-3.r-2.r-1.r+1}{120} - 2 \frac{r-3.r-2.r-1}{6} + 1 \frac{r-3}{1},$$

where it will be observed that we have the binomial coefficients 1, 2, 1 with the signs +, −, +.

*Three diagonals, B.*—There must be outside each of the three diagonals, that is, in three of the six intervals formed by the diagonals, two sides;

and there remain  $r - 6$  sides to be distributed between the six intervals; the number of ways in which this can be done is

$$= \frac{r - 5.r - 4.r - 3.r - 2.r - 1}{120},$$

and there is here no coalescence of diagonals, so that this is the number required.

19. *Four diagonals, A.*—There must be on each side of the four diagonals, that is, in two of the eight intervals formed by the diagonals, two sides; there remain

$r - 4$  sides to be distributed between the eight intervals, and the number of ways in which this can be done is

$$\frac{r - 3.r - 2.r - 1.r.r + 1.r + 2.r + 3}{5040}.$$

But this number requires to be corrected for coalescences, as in the case *Three diagonals, A*; and the required number is thus found to be

$$\begin{aligned} & \frac{r - 3.r - 2.r - 1.r.r + 1.r + 2.r + 3}{5040} - 3 \frac{r - 3.r - 2.r - 1.r.r + 1}{120} \\ & + 3 \frac{r - 3.r - 2.r - 1}{6} - 1 \frac{r - 3}{1}. \end{aligned}$$

*Four diagonals, B.*—There must be outside of three of the diagonals, that is, in each of three of the eight intervals formed by the diagonals, two sides; there remain  $r - 6$  sides to be distributed between the eight intervals, and the number of ways in which this can be done is

$$\frac{r - 5.r - 4.r - 3.r - 2.r - 1.r.r + 1}{5040}.$$

There is a correction for the coalescence of two of the diagonals, giving rise to a form such as *Three diagonals, B*; and consequently there is a term

$$- \frac{r - 5.r - 4.r - 3.r - 2.r - 1}{120},$$

which, with the first-mentioned term, gives the required number.

*Four diagonals, C.*—There must be outside of each of the diagonals, that is, in each of four of the eight intervals formed by the diagonals, two sides; there remain  $r - 8$  sides to be distributed between the eight intervals, and the number of ways in which this can be done is

$$\frac{r - 7.r - 6.r - 5.r - 4.r - 3.r - 2.r - 1}{5040},$$

which is the required number.

20. In the expressions of No. 14,  $A$ ,  $2A$ ,  $3A + 2B$ ,  $4A + 8B + 2C$ , if we regard the terminals of the diagonals as given points, then (1) we have two summits, which can be joined in one way only, giving rise to the diagonal-type  $A$ ; (2) we have four summits, which can be joined in two ways only, so as to give rise to the diagonal-type  $A$ ; (3) we have six summits, which can be joined in three ways so as to give rise to a diagonal-type  $A$ , and in two

ways so as to give rise to a diagonal-type  $B$ ; and (4) we have eight summits, which can be joined in four ways so as to give rise to a diagonal-type  $A$ , in eight ways so as to give rise to a diagonal-type  $B$ , and in two ways so as to give rise to a diagonal-type  $C$ ; we have thus the linear forms in question. To obtain the number of partitions, we have in each case to multiply by  $r$ . To explain this, after the polygon is drawn, imagine

C. XIII.

**14**